

ON AVAILABILITY IMPORTANCE OF COMPONENTS IN COHERENT SYSTEMS

B. T. Mangara

Department of Electronic, Electrical and Computer Engineering, Central University of Technology,
Free State (CUT), Private Bag X20539, Bloemfontein, 9300, South Africa
e-mail: aragnam@gmail.com

Key words: Availability; Reliability Importance; Performance; Cost-specific Importance.

Summary: The well-known Birnbaum's reliability measure of importance naturally defines the importance of a component in a coherent system of statistically independent, non-repairable components. By using a similar concept for repairable components, we discuss several availability measures of a component's importance. As availability of a component is defined by its times to failure and repair, the corresponding 'classical' measures of importance are modified to account for this case. Furthermore, costs that are needed to increase the time to failure and (or) to decrease the repair time are also taken into account. We analyse these measures and present several meaningful examples.

1. Introduction

In complex systems, it is necessary to translate system reliability requirements into detailed specifications for all components that make up a system. This process is often referred to as the Reliability Apportionment. During reliability apportionment, the reliability analyst has to understand and develop the relationships between component, subsystem, and system reliabilities. The crucial role in this process is in understanding and quantifying the reliability importance of different parts of the equipment.

The concept of *reliability* importance of components in a coherent system was proposed in the late 1960s (Birnbaum, 1969), although it is a specific case of a sensitivity analysis that was used in various engineering applications for ages. The well-known Birnbaum's reliability measure naturally defines the importance of the i^{th} component in a coherent system of n statistically independent, non-repairable components as (Birnbaum, 1969):

$$I_i^B = \frac{\partial h(\mathbf{P}(t))}{\partial p_i(t)} = h(1_i, \mathbf{P}(t)) - h(0_i, \mathbf{P}(t)), \quad (1)$$

where $h(\mathbf{P}(t)), t \geq 0$ is the reliability (survival) function of a system, $\mathbf{P}(t) = (p_1(t), p_2(t), \dots, p_n(t))$ is the vector of reliability functions of the components and $h(1_i, \mathbf{P}(t))$ ($h(0_i, \mathbf{P}(t))$), as usual, denotes the reliability function of a system with a given state of the i^{th} component at time t , that is,

$h(1_i \mathbf{P}(t)) = h(\mathbf{P}(t) \mid x_i(t) = 1) (h(0_i, \mathbf{P}(t)) = h(\mathbf{P}(t) \mid x_i(t) = 0))$, where $x_i(t)$ is a binary variable that indicates the state of component i for $i = 1, 2, \dots, n$. Equation (1) can be easily proved using the pivotal decomposition of the reliability function $h(\mathbf{P}(t))$ (Barlow and Proschan, 1975). Moreover, performing (sequentially) pivotal decomposition, one can eventually arrive at the following multi-linear representation, the importance of which (to the best of our knowledge) was underestimated in the literature. For brevity of notation, we will omit the time variable in what follows:

$$h(\mathbf{P}) = \sum_i^n c_i p_i + \sum_{1 \leq i < j \leq n} c_{i,j} p_i p_j + \sum_{1 \leq i < j < k \leq n} c_{i,j,k} p_i p_j p_k + c_{1,2,\dots,n} p_1 p_2 \dots p_n, \quad (2)$$

where coefficients $c_i, c_{i,j}, \dots$ can be obtained from the (similar to (2)) representation for the corresponding structure function of the system. In spite of its cumbersome form, (2) is methodologically convenient for qualitative analysis and for the interpretation of (1) and numerous other measures of importance (see, for example, Rausand and Hoyland (2004, pp. 183 - 206)) for measures which are popular in practice).

Denote by, $\phi(\mathbf{X})$, the structure function of our system, where $\mathbf{X} = (x_1, x_2, \dots, x_n)$ is the state vector with binary values of binary components ($x_i = 1$ when the i^{th} component is operable and $x_i = 0$ otherwise). Obviously, the similar to (2) multi-linear representation holds, where $h(\mathbf{P})$ is substituted by $\phi(\mathbf{X})$ and p_i by x_i , whereas the coefficients are the same. Note that this representation holds for systems with *dependent components* as well. However, as $h(\mathbf{P}) = E[\phi(\mathbf{X})]$, it is clearly seen now that (2) holds only for systems with statistically independent components (Rausand and Hoyland, 2004, pp. 148).

Assume that the reliability of the i^{th} component has been increased from p_i to $p_i + \Delta p_i \leq 1$ (all other components have not been changed). Then, from the multi-linearity of (2), it immediately follows that the change in reliability of the system can be written as:

$$\Delta_i h(\mathbf{P}) = \frac{\partial h(\mathbf{P})}{\partial p_i} \Delta p_i = I_i^B \Delta p_i. \quad (3)$$

Equation (3) is meaningful. When the i^{th} component has failed ($p_i = 0$) and is 'improved to the absolutely reliable state' then $\Delta p_i = 1$ and (3) reduces to (1). When $\Delta p_i = 1 - p_i$, we arrive at the measure that is usually called the *improvement potential* (Rausand and Hoyland, 2004, pp. 189 - 190):

$$\Delta_i h(\mathbf{P}) = \frac{\partial h(\mathbf{P})}{\partial p_i} (1 - p_i) = I_i^B (1 - p_i) = h(1_i, \mathbf{P}) - h(\mathbf{P}) \quad (4)$$

Another popular measure follows from (3) when $p_i = 0$ and $\Delta p_i = p_i$:

$$\delta_i h(\mathbf{P}) = \frac{\partial h(\mathbf{P})}{\partial p_i} p_i = I_i^B p_i = h(\mathbf{P}) - h(0_i, \mathbf{P}) \quad (5)$$

Hundreds of papers are devoted to measures of importance of non-repairable systems and only a few to repairable. However, there is one setting for repairable systems, where the measures of components' importance can be defined in a way similar to the non-repairable case. This is the case when a system's reliability characteristic of interest is its availability (see, for example, Barlow and Proschan (1975)). Thus, availability of the coherent system of n repairable components when each of

them is described by its own independent alternating renewal process can be defined as (Ross, 2003, pp. 586 - 587):

$$A(t) = h(\mathbf{A}(t)) = h(A_1(t), A_2(t), \dots, A_n(t)) \quad (6)$$

where $h(\cdot)$ is the *same* multi-linear function as in (1) – (4) with the substitution of arguments $p_i(t)$ by $A_i(t)$, where $A_i(t)$ is the availability of component i . However, in contrast to the non-repairable case, availability of a component is now defined by its *failure and repair times* and, therefore, the corresponding measures of importance should be modified to account for this. In section 2 we will suggest and analyse some measures of importance focusing on the stationary case (Rausand and Hoyland, 2004, pp. 371, 4: pp. 586 - 587):

$$A = \lim_{t \rightarrow \infty} A(t) = h(\mathbf{A}) = h(A_1, A_2, \dots, A_n) \quad (7)$$

What happens, if the components of our system are statistically dependent? This usually substantially complicates the corresponding general analytical reliability analysis and often makes it impossible. However, if we know the reliability (availability) of a system as a function of reliability (availability) functions of the components (or of the corresponding parameters), then, in principle, we can use the right-hand sides of equations (1) and (4)-(6) for obtaining different ‘direct’ measures of importance. Obviously, as was mentioned, the multi-linear representation (2) does not hold in this case and, therefore, the corresponding partial derivatives also do not make sense. We will also illustrate this reasoning in the next section using the corresponding example.

2. Measures of Availability Importance

Similar to (3), assume that (stationary) availability of the i^{th} component has been improved by ΔA_i , whereas other components are the same. Taking into account (6) and the corresponding multi-linear representation, it is easy to see that availability of our system has increased by (Barabady and Kumar, 2007):

$$\Delta_i h(\mathbf{A}) = \frac{\partial h(\mathbf{A})}{\partial A_i} \Delta A_i, \quad (8)$$

where notation $\Delta_i h(\mathbf{A})$ means that the corresponding increment $\Delta h(\mathbf{A})$ is due to the i^{th} component. Denote by T_i and τ_i the mean time to failure and the mean time of repair of the i^{th} component, respectively. Then, it is well-known that $A_i = T_i / (T_i + \tau_i)$, $i = 1, 2, \dots, n$. Equation (8) is a good indicator of importance for the i^{th} component, justifying in another way Birnbaum’s measure (1). Indeed, by increasing availability of each component by the same amount ΔA , we can choose the component with the maximum $\Delta_i h(\mathbf{A})$, which is the one with the maximum Birnbaum’s measure.

As stationary availability depends on two parameters (mean time to failure and mean time to repair), it is reasonable to define measures of importance with respect to each of them. The availability importance measure based, for example, on T_i (when $\tau_i, i = 1, 2, \dots, n$ is fixed) shows the effect of this parameter on availability of the whole system. Let:

$$I_{T_i} \equiv \frac{\partial h(\mathbf{A})}{\partial T_i} = \frac{\partial h(\mathbf{A})}{\partial A_i} \frac{\partial A_i}{\partial T_i} = \frac{\partial h(\mathbf{A})}{\partial A_i} \frac{\tau_i}{(T_i + \tau_i)^2} \quad (9)$$

Similarly, the availability importance measure based on τ_i (when $T_i, i = 1, 2, \dots, n$ is fixed) shows the effect of this parameter on availability of the whole system. Similar to (9), the availability importance measure based on τ_i can be calculated as follows:

$$I_{\tau_i} \equiv -\frac{\partial h(\mathbf{A})}{\partial \tau_i} = -\frac{\partial h(\mathbf{A})}{\partial A_i} \frac{\partial A_i}{\partial \tau_i} = \frac{\partial h(\mathbf{A})}{\partial A_i} \frac{T_i}{(T_i + \tau_i)^2}. \quad (10)$$

Note that, the function $h(\mathbf{A})$ is now non-linear either with respect to T_i or to τ_i . Therefore, it is interesting to compare the importance measure (9) with (8). Assume that T_i has been increased by $\Delta T_i \equiv \Delta T$ ($i = 1, \dots, n$), whereas τ_i has not been changed. Then (8) reads now:

$$I_{\Delta T, i} \equiv \Delta_i h(\mathbf{A}) = \frac{\partial h(\mathbf{A})}{\partial A_i} \frac{\Delta T \tau_i}{(T_i + \Delta T + \tau_i)(T_i + \tau_i)}. \quad (11)$$

Similarly, assume that τ_i has been decreased by $\Delta \tau_i \equiv \Delta \tau$ ($i = 1, \dots, n$) whereas T_i has not been changed. Then

$$I_{\Delta \tau, i} \equiv \Delta_i h(\mathbf{A}) = \frac{\partial h(\mathbf{A})}{\partial A_i} \frac{\Delta \tau T_i}{(T_i + \tau_i - \Delta \tau)(T_i + \tau_i)}. \quad (12)$$

Clearly, when ΔT ($\Delta \tau$) = $o(\Delta T$ ($\Delta \tau$)), (8) is close to (11) ((12)), that is,

$$\begin{aligned} I_{\Delta T, i} &= I_{T_i} [\Delta T + o(\Delta T)] \text{ and,} \\ I_{\Delta \tau, i} &= I_{\tau_i} [\Delta \tau + o(\Delta \tau)], \end{aligned}$$

where $o(\Delta T$ ($\Delta \tau$)) is called “ o small of delta T (τ)” and means: $\lim_{\Delta T (\Delta \tau) \rightarrow 0} o(\Delta T$ ($\Delta \tau$)) / ΔT ($\Delta \tau$) = 0.

However, if ΔT ($\Delta \tau$) $\neq o(\Delta T$ ($\Delta \tau$)) then, obviously, measures (11) and (12) should be used. The result of comparison of importance measures in this case can be different from that for small ΔT ($\Delta \tau$), which is shown by the following example.

Example 1. Consider a series system of two independent, repairable components (Ross, 2003; Barabady and Kumar, 2007; Pham, 2003):

$$A = h(\mathbf{A}) = A_1 A_2 = \frac{T_1}{T_1 + \tau_1} \frac{T_2}{T_2 + \tau_2};$$

In accordance with (11),

$$\begin{aligned} I_{\Delta T, 1} &= \frac{T_2}{T_2 + \tau_2} \frac{\Delta T \tau_1}{(T_1 + \Delta T + \tau_1)(T_1 + \tau_1)} \text{ and} \\ I_{\Delta T, 2} &= \frac{T_1}{T_1 + \tau_1} \frac{\Delta T \tau_2}{(T_2 + \Delta T + \tau_2)(T_2 + \tau_2)}. \end{aligned}$$

Assume that $\tau_1 \ll T_1; \tau_2 \ll T_2$, which is usually the case for repairable systems. Let $\Delta T \ll \min(T_1, T_2)$. Then,

$$I_{\Delta T, 1} \approx \Delta T \frac{\tau_1}{T_1^2} \text{ and } I_{\Delta T, 2} \approx \Delta T \frac{\tau_2}{T_2^2}.$$

If we now assume that ΔT is small we get the following:

$$I_{\Delta T, 1} \approx \Delta T \frac{\tau_1}{(T_1 + \Delta T) T_1}; \text{ and } I_{\Delta T, 2} \approx \Delta T \frac{\tau_2}{(T_2 + \Delta T) T_2}.$$

Let ΔT be sufficiently small and the importance of the first component be, for example, larger than that of the second one. However, when ΔT is not small, the result of comparison can be different. This can be illustrated by the following values of parameters:

$$T_1 = 1, T_2 = 1.5, \Delta T = 2, \tau_1 = 0.5\tau_2.$$

Example 2. Consider the series system of two repairable components when the whole system is switched-off during repair. In this case the components are dependent (via the repair actions) and availability of the system cannot be written as a product of availabilities of components. However, it can be obtained directly as (Pham, 2003):

$$A(T_1, T_2, \tau_1, \tau_2) = \frac{1}{1 + \frac{\tau_1}{T_1} + \frac{\tau_2}{T_2}};$$

Note that (8) does not hold now, but the direct sensitivity-type measures can be obtained as:

$$\begin{aligned} \frac{\partial A(T_1, T_2, \tau_1, \tau_2)}{\partial T_1} &= \frac{\tau_1}{T_1^2} \times \frac{1}{\left[1 + \frac{\tau_1}{T_1} + \frac{\tau_2}{T_2}\right]^2} = \tau_1 T_1^{-2} [A(T_1, T_2, \tau_1, \tau_2)]^2, \\ \frac{\partial A(T_1, T_2, \tau_1, \tau_2)}{\partial T_2} &= \frac{\tau_2}{T_2^2} \times \frac{1}{\left[1 + \frac{\tau_1}{T_1} + \frac{\tau_2}{T_2}\right]^2} = \tau_2 T_2^{-2} [A(T_1, T_2, \tau_1, \tau_2)]^2, \\ -\frac{\partial A(T_1, T_2, \tau_1, \tau_2)}{\partial \tau_1} &= \frac{1}{T_1} \times \frac{1}{\left[1 + \frac{\tau_1}{T_1} + \frac{\tau_2}{T_2}\right]^2} = T_1^{-1} [A(T_1, T_2, \tau_1, \tau_2)]^2, \\ -\frac{\partial A(T_1, T_2, \tau_1, \tau_2)}{\partial \tau_2} &= \frac{1}{T_2} \times \frac{1}{\left[1 + \frac{\tau_1}{T_1} + \frac{\tau_2}{T_2}\right]^2} = T_2^{-1} [A(T_1, T_2, \tau_1, \tau_2)]^2. \end{aligned}$$

From the above partial derivatives, it can be seen that all four measures have the common multiplier $[A(T_1, T_2, \tau_1, \tau_2)]^2$. The availability importance measure based on $T_i, i = 1, 2$ shows that the availability of the whole system is inversely proportional to the square of T_i . Hence, e.g., doubling (halving) T_i increases (decreases) availability of the whole system four times. The component with the largest value of $T_i (i = 1, 2,)$ has the smallest effect on the availability of the whole system.

On the other hand, the availability importance measure based on τ_i (the mean time of repair) of component i shows that the availability of the whole system is inversely proportional to T_i . The component with the largest value of T_i has the smallest effect on the availability of the whole system.

Remark 1. The direct sensitivity-type measures can also be obtained; e.g., for Example 1 (when

the whole system is not switched-off during repair) as:

$$\begin{aligned}\frac{\partial A(T_1, T_2, \tau_1, \tau_2)}{\partial T_1} &= \frac{T_2 \tau_1}{(T_1 + \tau_1)^2 (T_2 + \tau_2)}, \\ \frac{\partial A(T_1, T_2, \tau_1, \tau_2)}{\partial T_2} &= \frac{T_1 \tau_2}{(T_1 + \tau_1) (T_2 + \tau_2)^2}, \\ -\frac{\partial A(T_1, T_2, \tau_1, \tau_2)}{\partial \tau_1} &= \frac{T_1 T_2}{(T_1 + \tau_1)^2 (T_2 + \tau_2)}, \\ -\frac{\partial A(T_1, T_2, \tau_1, \tau_2)}{\partial \tau_2} &= \frac{T_1 T_2}{(T_1 + \tau_1) (T_2 + \tau_2)^2}.\end{aligned}$$

Assume that $\tau_1 \ll T_1; \tau_2 \ll T_2$, which is usually the case for repairable systems. The availability importance measure based on T_i (the mean time to failure) of component i shows that availability of the whole system is approximately τ_i/T_i^2 . Similarly, the availability importance measure based on τ_i (the mean time of repair) of component i shows that availability of the whole system is approximately $1/T_i$. Therefore, it can be seen from Example 1 and Example 2 that switching off the whole system during the repair introduces the common multiplier $[A(T_1, T_2, \tau_1, \tau_2)]^2$ to the direct sensitivity-type measures. Therefore, the availability importance measures have the same ordering for both cases (as it should be for the sufficiently “quick” repair).

3. Cost-based availability analysis

Consider availabilities of components of a coherent system, $A_i = T_i / (T_i + \tau_i)$, $i = 1, 2, \dots, n$ taking into account the costs that are needed to increase T_i on ΔT_i and to decrease τ_i on $\Delta \tau_i$. There are various ways to obtain the same increment in availability of the system (Finkelstein, 2008). The question is how to select the best, optimal in some sense way. For this purpose one should consider the costs necessary to achieve the same ΔA . In what follows, we discuss the heuristic approach to this problem. This approach is based on the general relation (8).

Define the cost-specific importance of the i^{th} component when availability of this component increases by a fixed amount $\Delta A_i \equiv \Delta A$, which is independent of i , as (Barabady and Kumar, 2007):

$$I_{c,j} = \frac{\Delta_j h(\mathbf{A})}{c_i(\Delta A, A_i)} = \frac{\Delta h(\mathbf{A})}{\Delta A_i} \frac{\Delta A}{c_i(\Delta A, A_i)} \quad (13)$$

where $c_i(\Delta A, A_i)$ - is the *minimum* cost necessary to improve A_i to $A_i + \Delta A$. When ΔA is reasonably small, which is usually the case, we can approximate $\frac{\Delta h(\mathbf{A})}{\Delta A_i}$ by the partial derivative, $\frac{\partial h(\mathbf{A})}{\partial A_i}$.

First, we must define the cost function $c_i(T_i, \tau_i)$ - the cost of the component with an average recovery time, τ_i and the average time to failure, T_i . Assume that it has the following additive form:

$$c_i(T_i, \tau_i) = F(T_i) + G(\tau_i) \quad (14)$$

where $F(x)$ and $G(x)$ - are monotonically increasing and monotonically decreasing functions, of its arguments, respectively. Thus $G(\tau_i)$, for example, shows that the cost is increasing when the time of repair is decreasing.

Using equation (14), we first define $c_i(A_i)$ - the minimum cost necessary to achieve the given value of A_i . Due to our assumptions on $F(x)$ and $G(x)$, and taking into account the condition $A_i = T_i / (T_i + \tau_i)$, the minimum always exists. Using this condition we first obtain, for example, the value of τ_i that results in the minimum of the corresponding univariate function of τ_i and then use $T_i = \tau_i \frac{A_i}{1-A_i}$ for the other variable. Suppose that this minimum is achieved with values of T_i^* and τ_i^* . Then $c_i(A_i) = c_i(T_i^*, \tau_i^*)$. In a similar way we define $c_i(A_i + \Delta A)$. Let this minimum cost be realized for values of T_i^{**} and τ_i^{**} , respectively. Then $c_i(A_i + \Delta A) = c_i(T_i^{**}, \tau_i^{**})$ and the minimum cost to improve A_i to $A_i + \Delta A$ is defined as

$$\Delta c_i(\Delta A) \equiv c_i(T_i^{**}, \tau_i^{**}) - c_i(T_i^*, \tau_i^*).$$

Example 3: Let $c_i(T_i, \tau_i) = a_i T_i + \frac{b_i}{\tau_i}$, where a_i and b_i - are constants. Then $\tau_i^* = \sqrt{\frac{b_i(1-A_i)}{a_i A_i}}$; and $T_i^* = \tau_i^* \frac{A_i}{1-A_i}$; $\tau_i^{**} = \sqrt{\frac{b_i(1-A_i-\Delta A)}{a_i(A_i+\Delta A)}}$; and $T_i^{**} = \tau_i^{**} \frac{A_i+\Delta A}{1-A_i-\Delta A}$.

Finally,

$$\Delta c_i(\Delta A) = a_i(T_i^{**} - T_i^*) + b_i \left(\frac{1}{\tau_i^{**}} - \frac{1}{\tau_i^*} \right). \quad (15)$$

Thus, expression (15) defines minimal $\Delta c_i(\Delta A)$ when A_i is realized in an optimal in the defined sense way. However, T_i and τ_i often correspond already to a 'given component', and, therefore, are not optimal. Nevertheless availability can be improved in an optimal way. In this case, one should find the minimum of

$$\Delta c_i(\Delta A) = c_i(T_i + \Delta T_i, \tau_i - \Delta \tau_i) - c_i(T_i, \tau_i);$$

given

$$\Delta A = \frac{T_i + \Delta T_i}{T_i + \Delta T_i + \tau_i - \Delta \tau_i} - \frac{T_i}{T_i + \tau_i}. \quad (16)$$

From (16), the ΔT_i can be expressed via $\Delta \tau_i$ and, therefore, the problem reduces to finding the minimum of the function of one variable. Thus, the procedure simplifies to finding the values of ΔT_i^* and $\Delta \tau_i^*$. Finally,

$$\Delta c_i(\Delta A) = c_i(T_i + \Delta T_i^*, \tau_i - \Delta \tau_i^*) - c_i(T_i, \tau_i).$$

Example 4: Combining examples 1 and 3 and assuming that A_1 and A_2 are realized with the optimal cost functions:

$$c_1(T_1^*, \tau_1^*) = a_1 T_1^* + \frac{b_1}{\tau_1^*} \quad (17)$$

and

$$c_2(T_2^*, \tau_2^*) = a_2 T_2^* + \frac{b_2}{\tau_2^*}. \quad (18)$$

Thus

$$\begin{aligned} c_1(A_1) &\equiv c_1 = 2\sqrt{a_1 b_1 \frac{A_1}{(1-A_1)}}, \quad c_2(A_2) \equiv c_2 = 2\sqrt{a_2 b_2 \frac{A_2}{(1-A_2)}}; \\ A_1(c_1) &\equiv A_1 = \frac{c_1^2}{4a_1 b_1 + c_1^2}, \quad A_2(c_2) \equiv A_2 = \frac{c_2^2}{4a_2 b_2 + c_2^2}. \end{aligned}$$

Using (13) and after simple algebraic transformations we finally arrive at the following quotient:

$$\frac{I_{c,1}}{I_{c,2}} = \frac{a_1 b_1 c_2 (4a_2 b_2 + c_2^2)}{a_2 b_2 c_1 (4a_1 b_1 + c_1^2)}.$$

Therefore, given the relevant parameters, the cost-specific importance measures for both components can be easily analysed and compared.

4. Conclusions

The well-known Birnbaum's reliability measure naturally defines the importance of the i^{th} component in a coherent system of n statistically independent, non-repairable components. It is obtained by partial differentiation of the system reliability with respect to that of component i . However, in contrast to the non-repairable case, availability of a component is defined by its time to failure and time of repair. When availability of a system is low, efforts are needed to improve it. In this study, two methods to improve availability of a repairable system are proposed. As stationary availability depends on two parameters (mean time to failure and mean time to repair) it is then reasonable to define measures of importance with respect to each of them. The availability importance measure based on T_i shows the effect of this parameter on the availability of the whole system. Similarly, the availability importance measure based on τ_i shows the effect of this parameter on the availability of the whole system.

Furthermore, to find the final strategy for the availability improvement process the cost trade-off is crucial. A cost-specific importance of component i when the availability of this component increases by a fixed amount is also proposed. To illustrate the proposed models, we have considered several meaningful examples.

If reliability of a system needs to be improved, then efforts should first be concentrated on improving reliability of the component that has the largest effect on reliability of a system. Similarly, the proposed availability importance measures can be used as a guideline for developing an improvement strategy in repairable systems.

References

- BARABADY, J. AND KUMAR, U. (2007). Availability allocation through importance measures. *International Journal of Quality & Reliability Management*, **24**, 643 – 657.
- BARLOW, R. E. AND PROSCHAN, F. (1975). *Statistical Theory of Reliability and Life Testing: Probability Models*. Holt, Rinehart, and Winston.
- BIRNBAUM, Z. W. (1969). On the importance of different components in a multi-component system. In KRISHNAIAH, P. R. (Editor) *Multivariate Analysis*. New York: Academic Press.
- FINKELSTEIN, M. (2008). *Failure Rate Modelling for Reliability and Risk*. Springer, London.
- PHAM, H. (2003). Commentary: Steady-state series-system availability. *IEEE Trans. Reliability*, **52** (2), 146 – 147.
- RAUSAND, M. AND HOYLAND, A. (2004). *System Reliability Theory: Models, Statistical Methods, and Applications*. 2nd edition. John Wiley, New York.
- ROSS, S. M. (2003). *Introduction to Probability Models*. 8th edition. New York: Academic.